INCH-POUND
MIL-I-24768/10
25 February 1992
SUPERSEDING
MIL-P-79C
15 June 1961

MILITARY SPECIFICATION SHEET

INSULATION, PLASTIC, LAMINATED, THERMOSETTING, PAPER-BASE, PHENOLIC-RESIN (PBE)

This specification is approved for use by all departments and agencies of the Department of Defense.

The requirements for acquiring the insulation material described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-I-24768.

REQUIREMENTS:

Construction: The material shall consist of a cellulosic substrate paper filler properly impregnated and bonded with a phenolic-resin matrix or binder and processed to conform to this specification.

Intended use: The laminated material covered by this specification is a moisture and humidity resistant plastic laminate designed for mechanical support in electrical applications. It possesses high dielectric strength and relatively low dielectric losses under conditions of adverse humidity and temperature.

Forms: Sheets, rolled tubes, molded tubes, and rods are recommended forms of laminated material covered by this specification.

Temperature indice: The laminated material covered by this specification has a mechanical strength test index of 130 and an electrical strength test index of 130 for thicknesses up to 0.062 inch, and a mechanical strength test index of 140 and an electrical strength test index of 140 for thicknesses greater than 0.062 inch.

Dimensions and part numbers:

Thickness of sheets: The thickness of laminated sheets, permissible variations, and the applicable part number shall be as specified in table I.

AMSC N/A FSC 5970
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202610-20T01:58Z
Check the source to verify that this is the current version before use.

TABLE I. *Thickness of laminated sheets.*

Part number	Nominal thickness (inches)	Permissible variations (± inch)
M24768/10-S-1	0.010	0.002
M24768/10-S-2	0.015	0.0025
M24768/10-S-3	0.020	0.003
M24768/10-S-4	0.025	0.0035
M24768/10-S-5	0.031	0.0035
M24768/10-S-6	0.047	0.0045
M24768/10-S-7	0.062	0.005
M24768/10-S-8	0.094	0.007
M24768/10-S-9	0.125	0.008
M24768/10-S-10	0.156	0.009
M24768/10-S-11	0.188	0.010
M24768/10-S-12	0.219	0.011
M24768/10-S-13	0.250	0.012
M24768/10-S-14	0.312	0.014
M24768/10-S-15	0.375	0.017
M24768/10-S-16	0.438	0.019
M24768/10-S-17	0.500	0.021
M24768/10-S-18	0.625	0.024
M24768/10-S-19	0.750	0.027
M24768/10-S-20	0.875	0.030
M24768/10-S-21	1.000	0.033
M24768/10-S-22	1.125	0.035
M24768/10-S-23	1.250	0.037
M24768/10-S-24	1.375	0.039
M24768/10-S-25	1.500	0.041
M24768/10-S-26	1.625	0.043
M24768/10-S-27	1.750	0.045
M24768/10-S-28	1.875	0.047
M24768/10-S-29	2.000	0.049
M24768/10-S-30	2.125	0.052
M24768/10-S-31	2.500	0.056
M24768/10-S-32	2.750	0.061
M24768/10-S-33	3.000	0.065
M24768/10-S-34	3.500	0.073
M24768/10-S-35	4.000	0.082
M24768/10-S-36	4.500	0.090
M24768/10-S-37	5.000	0.095
M24768/10-S-38	5.500	0.105
M24768/10-S-39	6.000	0.115
M24768/10-S-40	6.500	0.120
M24768/10-S-41	7.000	0.130
M24768/10-S-42	7.500	0.140
M24768/10-S-43	8.000	0.145
M24768/10-S-44	8.500	0.155
M24768/10-S-45	9.000	0.160
M24768/10-S-46	9.500	0.170
M24768/10-S-47	10.000	0.180
M24768/10-S-48	_[1]	_[2]

See footnotes on next page.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:58Z
Check the source to verify that this is the current version before use.

[1]For part numbers of sheets not listed in this table, the thickness shall be included after the part number. Example of this for a 0.8125 inch thick sheet shall be: M24768/10-S-48-0.8125.
[2]On sheets of nominal thickness not listed in this table, the permissible variations shall be the same as for the next greater thickness.

Diameter of rolled round tubes: The range of sizes for rolled round tubes shall be as specified in table II. The inside diameter and outside diameter shall be included in the part number. An example of a part number for a rolled tube with an inside diameter of 0.250 inch and an outside diameter of 0.312 inch shall be: M24768/10-TRR-0.250/0.312. The wall thickness tolerances for rolled tubes with an inside diameter up to 4 inches shall be as specified in table III.

TABLE II. *Range of sizes for rolled round tubes.*

Inside diameter (inches)		Outside diameter (inches)		Wall thickness (inch)		Maximum ratio of wall thickness to inside diameter
Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	
0.250	8.000	0.312	10.000	0.031	1.000	1/2

TABLE III. *Rolled round tube wall thickness tolerances.*

Wall thickness (inch)	Tolerances (± inch)
0.010 to 0.015	0.003
0.016 to 0.030	0.005
0.031 to 0.061	0.006
0.062 to 0.124	0.007
0.125 to 0.249	0.009
0.25 to 0.500	0.011

Diameter of molded round tubes: The range of sizes for molded round tubes shall be as specified in table IV. The inside diameter and outside diameter shall be included in the part number. An example of a part number for a molded round tube with an inside diameter of 0,.125 inch and an outside diameter of 0.250 inch shall be M24768/10-TMR-0.125/0.250. The wall thickness tolerances for molded round tubes with an inside diameter up to 3.875 inches shall be as specified in table V.

Source: https://assist.dla.mil -- Downloaded: 202610-20T01:58Z
Check the source to verify that this is the currentversion before use.

TABLE IV. *Range of sizes for molded round tubes.*

Inside diameter (inches)		Outside diameter (inches)		Wall thickness (inch)		Maximum ratio of wall thickness to inside diameter
Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	
0.125	3.875	0.250	4.000	0.062	1.000	1/2

TABLE V. *Molded round tube wall thickness tolerances.*

Wall thickness (inch)	Tolerances (± inch)		
	Inside diameter (inches)		
	0.125–0.250	*0.251–0.500*	*0.501–Max.*
0.031 to 0.062	0.008	0.008	0.008
0.063 to 0.125	0.011	0.011	0.011
0.126 to 0.250		0.015	0.011
0.251 to 0.500			0.013

Diameter of rods: The range of sizes and tolerances for rods shall be as specified in table VI. The outside diameter of the rod shall be included in the part number. An example of a part number for a rod with an outside diameter of 0.156 inch shall be: M24768/10-R-0.156.

TABLE VI. *Size and tolerances for rods.*

Diameter (inches)	Tolerances (± inch)
0.250 to 1.999	0.005
2.000	0.008

Source: https://assist.dla.mil -- Downloaded: 2026l0-20T01:58Z
Check the source to verify that this is the currentversion before use.

Performance requirements: Performance requirements for laminated materials shall be as specified in tables VII, VIII, IX, X, and XI.

The following performance requirements do not apply to this plastic laminate:

a. Surface resistance
b. Volume resistivity
c. Silicone content
d. Arc resistance
e. Thermal endurance
f. Flammability
g. Tracking resistance.

Testing: The laminated material covered by this specification shall be tested under first article inspection requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:58Z
Check the source to verify that this is the current version before use.

TABLE VII. *Performance requirements for sheets.*

Requirement	Cond.	Test para.	Unit	Thickness (inches)[1]									
				0.031	0.062	0.094	0.125	0.188	0.250	0.500	0.750	1.000	1.001 – Max.
Dielectric breakdown voltage[2]	A D-48/50	4.6.4	Min. kV	50.0 6.0	50.0 6.0	50.0 6.0	50.0 6.0	50.0 6.0	50.0 6.0	50.0 6.0	50.0 6.0	50.0 6.0	40.0 4.0
Dielectric constant	A D-24/23	4.6.6	Max.	5.30 6.00	5.30 5.90	5.30 5.80	5.30 5.70	5.30 5.70	5.30 5.70	5.30 5.70	5.30 5.70	5.30 5.70	5.30 5.70
Dissipation factor	A D-24/23	4.6.6	Max.	0.038 0.048	0.038 0.047	0.038 0.046	0.038 0.045	0.038 0.044	0.038 0.043	0.038 0.043	0.038 0.043	0.038 0.043	0.038 0.043
Volume resistivity	C-96/35/90	4.6.7	Min. megohm cm										
Surface resistance	C-96/35/90	4.6.8	Min. megohm										
Arc resistance	D-48/50	4.6.9	Min. sec.										
Tracking resistance	A	4.6.10	Min. minutes										
Impact strength[2] Lengthwise: Crosswise:	E-48/50	4.6.11	Min. ft – lbs. per inch							0.40 0.35	0.40 0.35	0.40 0.35	0.40 0.35

See footnotes at the end of table.

Source: https://assist.dla.mil -- Downloaded: 202610-20T01:58Z
Check the source to verify that this is the current version before use.

TABLE VII. *Performance requirements for sheets – Continued.*

Requirement	Cond.	Test para.	Unit	Thickness (inches)[1]									
				0.031	0.062	0.094	0.125	0.188	0.250	0.500	0.750	1.000	1.001 – Max.
Flexural strength													
Lengthwise:	A	4.6.12	Min. psi	13,500	13,500	13,500	13,500	13,500	13,500	13,500	13,500	12,000	12,000
Crosswise:				11,800	11,800	11,800	11,800	11,800	11,800	11,800	11,800	10,600	10,600
Thermal endurance *Flexural* strength:		4.6.13	Min. psi										
Bonding strength[2]	A	4.6.15	Min. pounds							950	950	950	950
	D-48/50									700	700	700	700
Water absorption	D_1-24/23	4.6.17	Max. percent	2.10	1.40	1.10	0.95	0.70	0.60	0.45	0.40	0.35	0.35
Silicone content	E-168/185	4.6.18	Max. ppm										
Flame resistance		4.6.19											
Ignition time:	A		Min. sec.										
Burning time:			Max. sec.										

[1]For intermediate thicknesses, the value for the next smaller thickness shall apply.
[2]Maximum thickness tested shall be 2.000 inches.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:58Z
Check the source to verify that this is the current version before use.

TABLE VIII. *Performance requirements for rolled tubes.*

Requirement	Cond.	Test para.	Unit	Inside diameter (inches)	Wall thickness (inches)			
					0.031–0.062	0.063–0.094	0.095–0.125	0.126–0.188
Dielectric breakdown voltage	D-48/50	4.6.4	Min. kV	0.500 to 8.000	18.0	18.0	18.0	18.0
Dielectric strength	D-48/50	4.6.5	Min. volts per mil	0.250 to 8.000	225	250	250	250
Compressive strength[1]	A	4.6.14	Min. psi	0.250–0.499 0.500–8.000		10,000 13,000	10,000 13,000	10,000 13,000
Specific gravity	A	4.6.16	Min.	0.250–0.499 0.500–8.000	1.12 1.12	1.12 1.12	1.12 1.12	1.12 1.12
Water absorption	D_1-24/23	4.6.17	Max. percent	0.250–2.000 2.001–8.000	3.5 3.5	1.5 1.5	1.3 1.3	1.0 1.0

TABLE VIII. *Performance requirements for rolled tubes – Continued.*

Requirement	Cond.	Test para.	Unit	Inside diameter (inches)	Wall thickness (inches)			
					0.189–0.250	0.251–0.375	0.376–0.500	0.501–1.000
Dielectric breakdown voltage	D-48/50	4.6.4	Min. kV	0.500 to 8.000	18.0	18.0	18.0	
Dielectric strength	D-48/50	4.6.5	Min. volts per mil	0.250 to 8.000	250			
Compressive strength[1]	A	4.6.14	Min. psi	0.250–0.499 0.500–8.000	10,000 13,000	10,000 13,000	10,000 13,000	10,000 13,000
Specific gravity	A	4.6.16	Min.	0.250–0.499 0.500–8.000	1.12 1.12	1.12 1.12	1.12 1.12	1.12 1.12
Water absorption	D_1-24/23	4.6.17	Max. percent	0.250–2.000 2.001–8.000	0.8 0.8	0.6 0.6		

[1]Values apply to inside diameters of 0.250 inch and over and outside diameters of 2.000 inches and less.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:58Z
Check the source to verify that this is the current version before use.

TABLE IX. *Performance requirements for molded tubes.*

Requirement	Cond.	Test para.	Unit	Inside diameter (inches)	Wall thickness (inches)						
					0.062–0.093	0.094–0.124	0.125–0.187	0.188–0.249	0.250–0.499	0.500–0.999	1.000
Compressive strength	A	4.6.14	Min. psi	0.250 to 3.875	20,000	20,000	20,000	20,000	20,000	20,000	20,000
Specific gravity	A	4.6.16	Min.	0.125 to 3.875	1.22	1.22	1.22	1.22	1.22	1.22	1.22
Water absorption	D_1-24/23	4.6.17	Max. percent	0.125 to 3.875	1.4	1.2	1.1	1.0	0.9	0.8	0.7

TABLE X. *Dielectric strength requirements for molded tubes.*

Requirement	Cond.	Test para.	Unit	Inside diameter (inches)	Wall thickness (inches)			
					0.062	0.063–0.125	0.126–0.250	0.251–0.500
Dielectric strength	A	4.6.5	Min. volts per mil	0.125 – 3.875	300	220	150	110

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:58Z
Check the source to verify that this is the current version before use.

TABLE XI. *Performance requirements for rods.*

Requirement	Conditioning	Test para.	Unit	Diameter (inches)				
				0.125–0.249	0.250–0.499	0.500–0.999	1.000	1.001–2.000
Flexural strength	A	4.6.12	Min. psi	13,000	13,000	13,000	13,000	
Compressive strength	A	4.6.14	Min. psi	20,000	20,000	20,000	20,000	20,000
Specific gravity	A	4.6.16	Min.	1.25	1.25	1.25	1.25	1.25
Water absorption	D_1-24/23	4.6.17	Max. percent	1.5	1.0	0.75	0.75	0.75

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:58Z
Check the source to verify that this is the current version before use.

Custodians:
Army – ER
Navy – SH
Air Force – 20

Review activities:
Army – MI, AR, EA
Navy – EC, OS
Air Force – 85

User activities:
Army – ME
Navy – MC, AS

Preparing activity:
Navy – SH
(Project 5970-1015-10)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:58Z
Check the source to verify that this is the current version before use.